

ECE 647: Lecture 18

Sharing a Single Resource

- R : the amount of resource:
 - bandwidth
 - stocks, etc.
- N : number of users
- x_s : the amount of resource allocated to user s , where $s = 1, \dots, N$
 - Assume x_s is real
 - the resource is infinitely divisible

Clearly we have constraint

$$\sum_{s=1}^N x_s \leq R$$

How to allocate the resource among multiple uses?

- Equally?
- Who paid highest amount of money?

(a) Utility Maximization (Welfare Maximization)

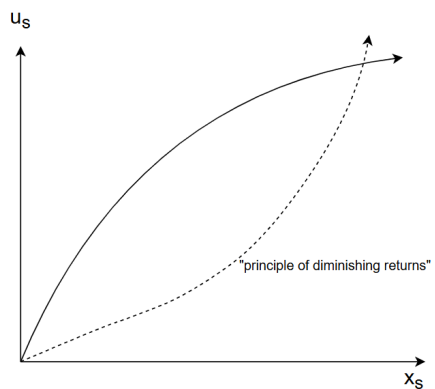
- $u_s(x_s)$: The utility to user s if they have x_s amount of resource.
- Maximize total utility:

$$\max \sum_{s=1}^N u_s(x_s)$$

subject to:

$$\sum_{s=1}^N x_s \leq R, \quad x_s \geq 0$$

- A convex problem if u_s 's are concave.



Fairness

- Concavity of utility function can be used to model fairness

Examples

(a) $u_s(x_s) = w_s \log(x_s)$

↑ Constant weight

- Proportional Fair
 - Let x_s^* be the optimal solution, then:

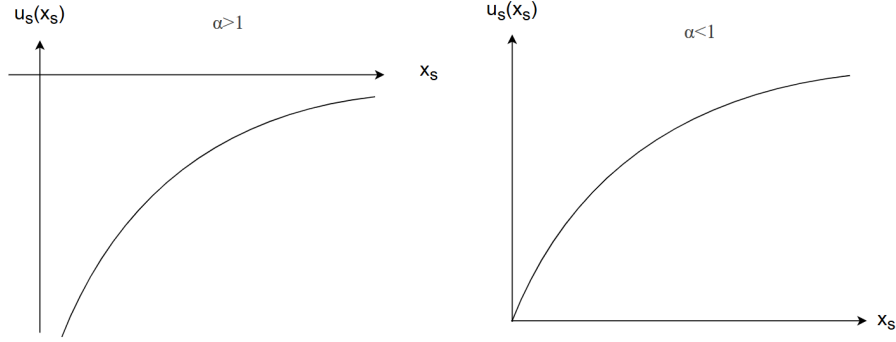
$$u'_s(x_s^*) = \frac{w_s}{x_s^*}$$

- For any other feasible allocation x_s , we must have:

$$f'(x^*; x - x^*) = \sum \left(\frac{w_s}{x_s^*} \right) (x - x_s^*) \leq 0$$

– The sum of relative rate differences cannot be improved.

(b) $u_s(x_s) = w_s \cdot \frac{x_s^{1-\alpha}}{1-\alpha}, \quad \alpha > 0 \quad (\alpha\text{-fairness})$



- As $\alpha \rightarrow 1$, $u_s(x_s) \rightarrow w_s \log(x_s)$.
- For any optimal x_s^* , we have:

$$u'_s(x_s^*) = \frac{w_s}{(x_s^*)^\alpha}$$

$$f'(x^*; x - x_s^*) = \sum_s \frac{w_s(x_s - x_s^*)}{(x_s^*)^\alpha} \leq 0$$

α -fairness.

- When $\alpha \rightarrow 0$:

$$\max_s \sum w_s x_s \Rightarrow \text{weighted sum-rate}$$

- What happens when $\alpha \rightarrow \infty$?

$$\max_s [\min x_s] \Rightarrow \text{max-min fairness}$$

Solution: Sharing for Single Resource

- When $u_s(\cdot)$ is increasing:

$$\Rightarrow \sum x_s = R$$

– Optimum is at the boundary.

- Assume further that x^* satisfies $x_s^* > 0$ for all s .
- Then the normal vector is simply $[1, \dots, 1]$. We must have:

$$-[-u'_s(x_s)] = \lambda \cdot 1 \quad \text{for all } s$$

- This can be thought of as each user maximizing:

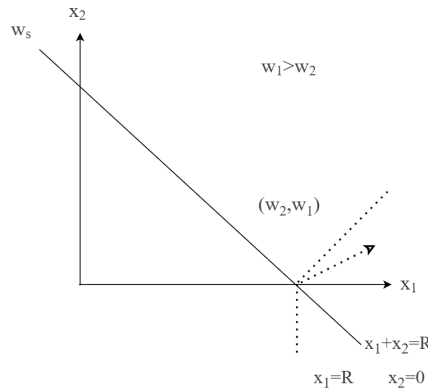
$$\max_{x_s} [u_s(x_s) - \lambda x_s]$$

where λ is the price.

- True even if $x_s^* = 0$ for some s .

What if $u_s(x_s) = w_s x_s$?

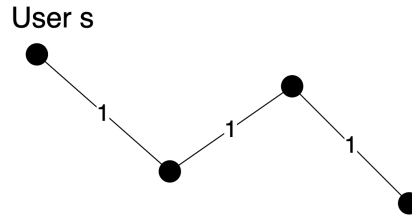
- The optimal solution always assigns R to the user with the largest w_s .



- Following the gradient direction of increasing $\sum_s u_s(x_s)$, the value of x_s will naturally go to: $x_1 = R, x_2 = 0$

Rate allocation of Internet: Multiple Resources

Single-Path



- R^l : capacity of link l
- x_s : rate allocated to user s
- $u_s(x_s)$: utility to user s
- $[H_s^l]$: routing matrix $H_s^l = \begin{cases} 1, & \text{if user } s \text{ uses link } l \\ 0, & \text{otherwise} \end{cases}$
- $\sum_s H_s^l x_s$ = total amount of traffic on link l

How to allocate the rates?

$$\max \sum_s u_s(x_s)$$

$$\text{s.t.} \quad \sum_s H_s^l x_s \leq R^l \quad \forall l$$

- Convex problem if $u_s(\cdot)$ is concave
- Physical meaning: Congestion control
 - High throughput
 - Avoid congestion
 - Fairness (related to utility function)
- Duality: Optimality condition

Multipath

- Jointly routing & controlling congestion.

$$H_{sj}^l = \begin{cases} 1 & \text{if path } j \text{ of user } s \text{ uses link } l \\ 0 & \text{otherwise} \end{cases}$$

- Let X_{sj} be the data rate of user s on path j .
 - **Constant:** $\sum_{sj} H_{sj}^l X_{sj} \leq R^l$
 - **Objective ?**